

SEAT No. _____



No. of Printed Pages: 2

[18/19]

SARDAR PATEL UNIVERSITY
B. Sc.(CA & IT) EXAMINATION, IV SEM

Date : 16th Apr 2022, Saturday
Session : **NOON**
Course No : PS04SIIT22

Time : 12:00pm To 02:00pm
Sub: Operation Research - II
Total marks : 35

Q – 1 Multiple Choice Question

[05]

- i) The _____ method used to obtain optimum solution of assignment problem.
a) Simplex b) Hungarian c) Dominance d) Graphical
- ii) The size of the payoff matrix of a game can be reduced by using the principle of _____.
a) Game Inversion b) Rotation Reduction
c) Dominance d) Game of Transpose
- iii) _____ operation is carried out on a machine at a time..
a) Only one b) two c) At least one d) None
- iv) Activity which starts only after finishing other activity is called _____.
a) Dummy b) Successor c) Predecessor d) None
- v) Burst and merge are type of _____ in Networking.
a) Event b) Activity c) Arrow d) Tools

Q – 2 Fill in the Blanks and True – False

[04]

- i) If number of source is not equal to number of destination in assignment problem then it is called _____.
- ii) Games which in value more than two players are called _____.
- iii) In sequencing if smallest time for a job belongs to machine – 1 then that job has to placed in the middle. (True / False)
- iv) Dummy activity is indicated by hardline arrow. (True / False)

Q – 3 Short Answer attempt any five (Each carry 2 marks)

[10]

- i) Write an algorithm to solve Assignment Problem
- ii) Write about Assignment Problem
- iii) Write characteristics of game theory
- iv) Define saddle point and value of game.
- v) Write in brief regarding problem of sequencing.
- vi) Write down any two assumptions used for solving sequencing problem.
- vii) Differentiate between system design and detailed design

Q – 4 Attempt any four (Each carry 4 marks)

[16]

- i) Write the steps for solving a A.P. by hungarian method..
- ii) State the rules for drawing network diagram
- iii) Write down the procedure to obtain optimum completion time using critical path method.

- iv) Solve the following Assignment Problem so as to minimize the time required to complete all the task

Job	Task				
	1	2	3	4	5
A	6	5	8	11	16
B	1	13	16	1	10
C	16	11	8	8	8
D	9	14	12	10	16
E	10	13	11	8	16

- v) Consider game matrix which represents payoff to the player A. Find the optimal strategy, if any.

	I	II	III	
I	-3	-2	-3	A
II	2	0	2	
III	5	-2	-4	

- vi) We have five jobs each of which must to go through two machines in the order A B. Processing Times are given in the table below:

Job	1	2	3	4	5
Machine A	10	2	18	6	20
Machine B	4	12	14	16	8

Determine a sequence for the five jobs that will minimize the total elapse time.

- vii) A small maintenance project consists of the following 12 jobs. Draw the arrow network of the project. Determine the critical path.

JOBS	Duration in Days	JOBS	Duration in Days
1-2	2	5-8	5
2-3	7	6-7	8
2-4	3	6-10	4
3-4	3	7-9	4
3-5	5	8-9	1
4-6	3	9-10	7

- viii) Find the critical path and calculate the Total float and Free float for the following PERT diagram.

